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MEMORANDUM FOR RECORD

16 September 1954

SUBJECT: Meeting of Upper Atmosphere Rocket Research Panel

1. At the invitation of Air Force Scientific Advisory Board, I attended a meeting of the Upper Atmosphere Rocket Research Panel on 8 September at the Naval Research Laboratory, Arlington. The two agenda items of primary concern were: 1) Higher Altitude and Satellite Vehicles, 2) International Geophysical Year, and the logistics therefor.

2. This panel was formed in 1948 by representatives of various organizations which were concerned with upper atmosphere research through rocketry. It is chaired by Dr. James A. Van Allen who is Chairman of the Department of Physics, State University of Iowa. The panel has representation from Aberdeen Proving Grounds, University of Michigan, Army Signal Laboratory, General Electric Company, Naval Research Laboratory, Air Force Cambridge Research Station, California Institute of Technology, and Harvard Observatory. It has no official connection with any department of defense or aviation but is informally supported by the Office of Naval Research. In connection with the forthcoming International Geophysical Year, it has been assigned the function of developing the United States program for upper atmosphere research through the use of high altitude rockets.

3. At the 8 September meeting, there were in addition to the panel itself, representatives from the Chief of Ordnance, United States Army, the Ordnance Arsenal, the Rand Corporation, the Office of Naval Research, Aerophysics Development Corporation, Air Force AGO, AFOS, National Science Foundation, etc.

4. Dr. Van Allen opened the discussion of the first agenda item - Higher Altitude and Satellite Vehicles - by a discussion of past upper atmosphere research with rockets. He cited the extensive use of V-2s which could carry 2,000 pounds payload to a 100 mile altitude but the supply of which has now been exhausted. He mentioned the WAC-Jupiter which was not used as a single stage vehicle, primarily because it could carry only 24 pounds to an altitude of 80 miles. The WAC-Jupiter was however, used as a second stage in combination with the V-2 and achieved an altitude of 200 miles. Since exhaustion of the V-2 supply, the Army rockets have been developed and used as civilian research rockets. These are the Viking which in its present form can carry 400 pounds payload to 135 miles

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altitude and in its forthcoming modification will carry 500 pounds to 125 miles altitude. The second rocket is the Aerobee which can carry 150 pounds payload to 65 miles altitude. In addition to these two, there is also the Wallops which is a system of launching a rocket rocket from a "balloon" balloon. The launch is made at a balloon altitude of approximately 100,000 feet and the rocket carries 30 pounds of instrumentation to an altitude of 60 miles. Dr. Van Allen concluded his presentation by stating that, from here on, civilian upper atmosphere rocket research will probably be nonexistent, due to lack of civilian funds, upon the military rockets being developed now by the Department of Defense. He then introduced Dr. Fred Hinkle of the Harvard Observatory for a discussion of Earth Satellite Vehicle (ESV).

5. Dr. Hinkle stated that the interests of civilian research and of the Department of Defense in ESV are synonymous. He pointed out the main scientific gains which could be made through the ESV which have application both to civilian science and to defense. These include meteorology, astronomy, solar-terrestrial relations, far ultra-violet and X-ray research, etc. He described the ESV sequence as being in three phases as follows:

(1) The placing in orbit of an observable object (uninstrumented) which could be seen either optically or by radar.

(2) The placing in orbit of an instrumented vehicle, "an unmanned physical laboratory". This phase will be a progressive development starting with a very small vehicle similar to the Wallops, which will carry telemetering equipment. Larger capacity will be achieved as bigger and more reliable power plants are developed. These latter will probably be nuclear or solar energy plants. Instrumentation starting with simple telemetering will progress through more complicated stages until television and finally a telescope is included. The last stage in this phase will be a remote controlled vehicle.

(3) The final phase of ESV will be the putting in orbit of a manned satellite vehicle.

6. Dr. Hinkle emphasized that the placing of a slug in orbit was in Phase One first step in the process. He stated that such a vehicle even without instrumentation could produce useful scientific results such as air density data and relative positions on the earth. The main problem in connection with a slug phase 1 is that of observation from the earth, which will require much study. This problem will be simplified if the ESV is in an either an equatorial or a polar orbit. In regard to

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the unmanned instrumented vehicle (Phase 2), he said that the main problems will be development of small reliable power plants; television, the technical design of which is already well along; orientation of the vehicle itself and of the instrumentation carried; and constant reduction of weight of the equipment to be carried. He stated that the altitude at which the vehicle should orbit will depend upon the purpose envisaged. Theoretically, an altitude of 1,000 miles at a speed of 5 miles per second would be ideal. This would provide a 2-hour orbit.

7. Mr. George Hoover of the Air Branch, GPR, made the next presentation, concerning high altitude vehicle projects with which GPR is concerned. GPR has three main projects in this field, the first of which is the development of a manned conventional aircraft to operate at a maximum altitude of 300 miles. Two designs are presently under consideration - the Douglas 553 and the Douglas 555. The former is designed to have an altitude of 700,000 feet. (Note: this field of study is covered in Report R-57-1266100, "High Altitude and High Speed Study" by Douglas Aircraft Corporation, Santa Monica, Calif.). The second GPR project is development of a manned high altitude balloon to operate at 100 to 200,000 feet. This system would be based on the "Skyhook" polyethylene balloon carrying a gondola equipped to sustain one or two men. The third GPR project is for manned space flight and the study of this is being conducted at the Aero Jet Laboratory. The first phase of this latter project is called SMOG, the purpose of which is to place an RV in orbit at an altitude of 200 miles in order to secure meteorological and atmospheric data at that level as the first step toward higher altitude work. Project SMOG has been tentatively approved by the Navy and GPR is going ahead with it in cooperation with the Army. It is now also being coordinated with the Air Force at a very high level. The project calls for the use of the Army Redstone missile (see para. 3 below) as the first stage with the Loki cluster (see para. 9 below) providing the second and third stages. Under Aero Jet leadership, four subsidiary studies are planned to be undertaken as follows:

(1) A Feasibility Study to determine the size and weight of the vehicle required at an altitude of 300 miles. It is hoped that Mr. Fred Whipple will be the leader of this.

(2) An orbital Study to determine the power required, the guidance system, etc. It is hoped that Mr. B. Fred Singer of the University of Maryland will lead this.

(3) A Reentry-trajectory Study to determine the final design and the staging requirements.

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(b) A launching study to determine where and how the vehicle should be launched, the logistics requirements, and the range risks involved.

Following the completion of these four studies, construction of the actual vehicle will be commenced. All considers that successful completion of Project 1111 will lead into the launching of a vehicle similar to the 1111, i.e., an instrumented vehicle using a polar orbit at an altitude of 200 miles. It is expected that 1111 would remain aloft for ten days while 1111 could probably remain in its orbit for about one month. Mr. Mr. Hoover emphasized that if adequate backing is available for Project 1111, 1111 right well be used during the International Geophysical Year. He laid great stress on the necessity for the United States being the first in launching an 1111 and said that Project 1111 was absolutely essential to achieve this end.

8. Mr. Miller of Rockstone Arsenal then gave a briefing on the Rockstone Arsenal. The Rockstone Arsenal Model 1111-11, as a tactical weapon, has a range of 140 miles with a trajectory apex of 11 miles. If used as a research tool, it has a vertical range of 110 miles with a weight of about 10 pounds. The modified 1111 which is now under development will have a vertical range of 270 miles with an altitude of about 100 miles. Rockstone Arsenal is now working on the construction of these missiles as research instruments. The cost will cost 70,000 to 100,000 per missile if more than five missiles are produced. They have a speed of approximately 1,000 m.p.h.

9. Mr. Miller of Rockstone Arsenal described the Rockstone Arsenal vehicle which they are developing as an air borne project. The vehicle carries two clusters of 1111 rockets; the first cluster of seven acting as the first stage booster; the second cluster of four constituting the second stage. It has a speed of about 10 and a payload of 10 pounds. Instrumented vertically with an altitude of 10,000 feet, it is designed to operate at an altitude of 100 miles. It is planned to use either a flashing light system or a following radar reflector for tracking purposes. Instrumented, it is expected to sit at an altitude of 200 miles. After the first stage, the second stage develops 100 m.p.h. The missile has yet to be flown vertically as a research instrument. Mr. Miller indicated that the second stage cluster could be adapted to carry 12 pounds of ruggedized tolerating equipment.

10. Mr. Miller of Rockstone Arsenal commented briefly on a design which they are developing. It consists of the 1111 booster as the first stage and a seven rocket as the second stage. It is estimated that this vehicle could attain an altitude of 100,000 feet. It would be very low cost, in the range of \$7,000 to \$8,000.

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11. Mr. William of the National Science Foundation gave a brief account of the amount of the funds for upper atmosphere rocket research. One million, three-hundred thousand dollars is presently available and about 70% of this will be transferred immediately to ONR for procurement and accounting purposes. The balance of the rocket project funds - \$400,000 - will become available next year.

12. Following adjournment of the panel meeting, I spent a couple of hours with Mr. Van Allen, who had served under me for a time during World War II. Van Allen was one of the key figures in the development of the W-2 fuse under Admiral Parsons and was one of the officers assigned to introduce the fuse to the Pacific Fleet. In our conversation, I mentioned the difficulty that apparently would be encountered in the W-2 program, of equipping scientific equipment which would stand the high G's encountered in most of the cluster rockets, particularly with solid propellant. I mentioned specifically the G's encountered in stage two of the Loki cluster vehicle. Mr. Van Allen said that there should be no such difficulty. He pointed out that the W-2 fuse contained five miniature radio tubes, a battery and a transmitter, and that this instrument, with practically no failures, withstood 20,000 G's when fired from the Loki 4-inch anti-aircraft gun. He pointed out also that in the Loki 4-inch anti-aircraft gun, in use for some time, the Loki fuse at various points of instrumentation for scientific research had withstood and withstood 30 G's. He said that he saw no difficulty whatsoever in equipping instrumentation for high altitude rockets which would easily withstand 1,000 G's. In regard to the W-2 program generally, he agreed thoroughly with Mr. Van Allen, that the program must step up to launch a ship. He felt that if top level government priority could be established, with guidance and control by an outside high level scientific group, it would almost certainly be possible to put a ship in orbit by the time of the W-2 and possibly even put up an instrumented vehicle.

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